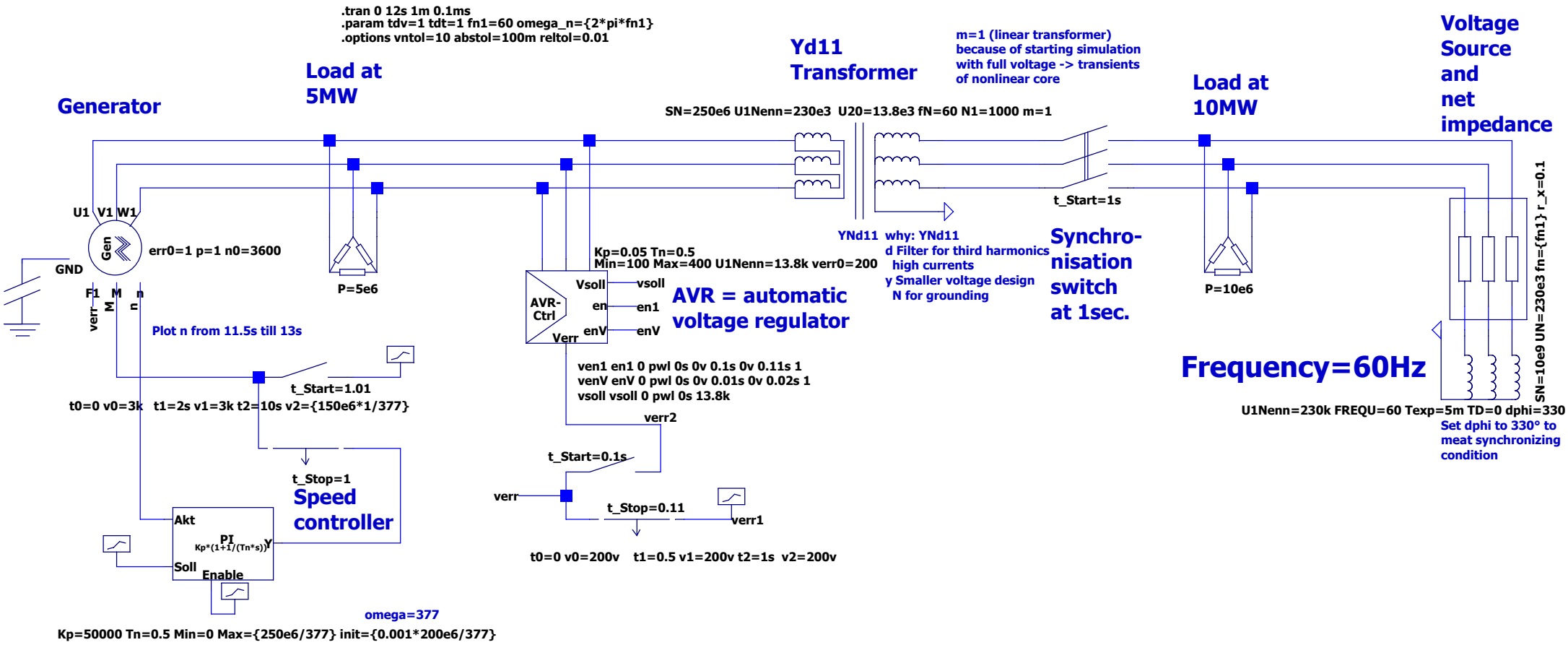
.tran 0 12s 1m 0.1ms
.param tdv=1 tdt=1 fn1=60 omega_n={2*pi*fn1}
.options vntol=10 abstol=100m reltol=0.01
Generator
Load at 5MW
Yd11 Transformer
m=1 (linear transformer) because of starting simulation with full voltage -> transients of nonlinear core
Load at 10MW
Voltage Source and net impedance
SN=250e6 U1Nenn=230e3 U20=13.8e3 fN=60 N1=1000 m=1
t_Start=1s
U1 V1 W1
Gen
GND
err0=1 p=1 n0=3600
F1 M n
verr
M
n
P=5e6
YNd11
why: YNd11
d Filter for third harmonics high currents
y Smaller voltage design
N for grounding
Synchro-nisation switch at 1sec.
P=10e6
SN=10e9 UN=230e3 fn={fn1} r_x=0.1
Kp=0.05 Tn=0.5
Min=100 Max=400 U1Nenn=13.8k verr0=200
Vsoll
en
enV
Verr
vsoll
en1
enV
AVR-Ctrl
AVR = automatic voltage regulator
Plot n from 11.5s till 13s
t_Start=1.01
t0=0 v0=3k t1=2s v1=3k t2=10s v2={150e6*1/377}
ven1 en1 0 pwl 0s 0v 0.1s 0v 0.11s 1
venV enV 0 pwl 0s 0v 0.01s 0v 0.02s 1
vsoll vsoll 0 pwl 0s 13.8k
verr2
t_Start=0.1s
verr
t_Stop=0.11
verr1
t0=0 v0=200v t1=0.5 v1=200v t2=1s v2=200v
Frequency=60Hz
U1Nenn=230k FREQU=60 Texp=5m TD=0 dphi=330
Set dphi to 330° to meat synchronizing condition
t_Stop=1
Speed controller
Akt
PI
Kp*(1+1/(Tn*s))
Y
Soll
Enable
omega=377
Kp=50000 Tn=0.5 Min=0 Max={250e6/377} init={0.001*200e6/377}